NHH

TERM PAPER
FIE463

Spring, 2026

Start: Thursday, April 9, 09:00

End: Wednesday, April 22, 12:00

THE TERM PAPER SHOULD BE SUBMITTED IN WISEFLOW AND CANVAS

You can find information on how to submit your paper here:
https://www.nhh.no/en/for-students/examinations/home-exams-and-assignments/

Your candidate number will be announced on StudentWeb. The candidate number should be noted on all pages (not your name or student number). In case of group examinations, the candidate numbers of all group members should be noted.

Collaboration between individuals or groups on submission preparation, as well as exchange of self-produced materials between individuals or groups is prohibited. The answer paper must consist of individual's or the group's own assessments and analysis. All communication during the home exam is considered cheating. All submitted assignments are processed in Ouriginal, a plagiarism control system used by NHH. Use of artificial intelligence (AI) in this exam is determined by the course coordinator. Information on AI usage and guidelines can be found on the course's Canvas page.

SUPPLEMENTARY REGULATIONS FOR EXAMINATIONS

You can find supplementary regulations under the headline “Regulations”

https://www.nhh.no/en/for-students/regulations/

Find more information under chapter 4.0 in the Supplementary provisions to the regulations for fulltime study programmes

Number of pages, including front page: 11

Number of attachments: 1

Group term paper 2

FIE463: Numerical Methods in Macroeconomics and Finance using Python

About the group term paper

- Deadline: Wednesday, April 22 at 12:00 (noon). Late submissions will not be accepted.
- Your solution needs to be submitted **both on Wiseflow** (for grading) **and on Canvas** (for peer review).
- Submissions must be made in groups of 2–3 students. Cooperation across groups is not allowed.
- You are allowed to use AI for support, but you are not allowed to hand in output produced directly by AI as your own solution (see AI Regulations on the Canvas course page).

Requirements

- Your solution must be implemented in a *single* Jupyter notebook.
- You need to include all *additional* data files that you have created in the `data/` folder. You don't need to include the original CSV files.
- You should submit a ZIP file containing both your Jupyter notebook and the additional data files. The ZIP file should contain the following files and folders:

```
ZIP file
|-- solution.ipynb
`-- data/
    |-- file1.csv
    |-- ...
    `-- fileN.csv
```

- Make sure your notebook runs without errors (`Restart` and `Run All`).
- Your implementation *must* work with the *FIE463* Anaconda environment we created in the course.
- Your notebook should be well-structured and visually appealing, e.g.,:
  - Use markdown cells with headings and other formatting elements (see here and here).
  - Each part should be in a clearly distinguishable subsection.
  - Each function should be defined in its own code cell.
  - Each graph should be in its own cell.
  - Graphs should be visually appealing. They should include legends, labels, and other annotations so they are easy to comprehend.
  - Don't print hundreds of lines of output in your final notebook. This is particularly relevant for `scikit-learn` since some models can generate numerous warnings when called with the wrong arguments.
- Your code should satisfy the following criteria:

- Write code that is reusable and avoid duplication. An excellent submission should contain only one implementation of each function that is called repeatedly to solve different tasks.
- Your code should be efficient. For example, use vectorization within NumPy or Pandas instead of loops where it makes sense.
- Your code should be commented to help the reader understand what it is doing.
- Your code should be well-formatted. You can use the Ruff formatter extension for VS Code for this purpose.

- Be sure to provide intuitive explanations for your results. These explanations should be *concise* and *must* be provided as markdown text, *not* as Python code comments.

Assessment

Your term paper will be assessed according to the following criteria:

Component	Points
Solution for Part 1	10
Solution for Part 2	10
Solution for Part 3	10
Solution for Part 4	10
Solution for Part 5	30
Solution for Part 6	25
Solution for Part 7	5

Note that a correct result is not sufficient to obtain full points: code efficiency, code reuse, elegance of the solution, and the presentation of the results are also factored in. The term paper contains a few optional parts, such as fitting additional models that we have not covered in the course. These optional tasks are not required to receive full points.

Please read the section on Performance issues **before** starting.

Expectations about macroeconomic and financial variables

Introduction

In this term paper you are asked to build a model to predict household expectations about macroeconomic and financial variables such as inflation, the unemployment rate, house prices, and the stock market. This is a hot topic in the literature using machine learning and LLMs to answer economic and financial questions (see, for example, Zarifhonarvar, A. (2025). *Generating inflation expectations with large language models*. Journal of Monetary Economics; Hansen, A. L., Horton, J. J., Kazinnik, S., Puzzello, D., & Zarifhonarvar, A. (2026). *Simulating the survey of professional forecasters*; Wu, J. C., Xi, J., & Xie, S. (2025). *LLM Survey Framework: Coverage, Reasoning, Dynamics, Identification*). It is also of great interest to policymakers, since expectations about inflation, the likelihood of becoming unemployed, or the profitability of the stock market are important drivers of household choices. Policymakers thus often try to directly influence such expectations, for example by using forward guidance and inflation targets to anchor inflation expectations. Being able to model expectations allows policymakers to assess potential changes in household behavior without having to rely on direct survey evidence, which is usually only available with some delay.

Whereas the papers listed above use LLMs or other more complex methods to model expectations, you will be employing `scikit-learn` for this purpose. The data for this task comes from the Survey of Consumer Expectations (SCE), a monthly online survey of approximately 1,200 respondents that has

been running since 2013 and is administered by the Federal Reserve Bank of New York, a branch of the US central bank.

In addition, you will be augmenting this dataset with macroeconomic and financial variables from Yahoo! Finance and FRED.

Your task is to train models on the survey data up to the end of 2023 and use these models to predict expectations in 2024. Unlike the first term paper, fewer detailed instructions are provided, and you are free to approach this task in whichever way you want (this includes using models in `scikit-learn` that we have not covered in the course).

Individual-level SCE data

The SCE raw data is distributed as three large and poorly structured Excel files, but you will be working with a subset of the data which has already been preprocessed and made available in CSV format by survey year. The appendix to this term paper provides a detailed list of variables.

The individual-level SCE data for this term paper are stored in several files in the folder `data/`, which have filenames following the pattern `sce_extract_YYYY.csv`, where `YYYY` is the year in which a particular wave of the survey was run. Each line in these files corresponds to an individual-wave observation in that year.

The SCE is a rotating panel: individuals are recruited into the survey and usually remain for 12 months (yielding 12 observations per individual) and are then replaced with new respondents. A handful of individuals are observed more than 12 times, while others stop responding before the 12th month.

Each individual is assigned a unique identifier (variable `userid`), which remains constant across all survey waves in which they are observed. Each survey wave is also assigned a unique identifier (variable `wid`).

Note that some variables (such as homeownership status, self-reported health, and the numerical literacy questions) are asked only the first time an individual answers the survey and are missing in all subsequent waves.

For this term paper, you can make the following simplifying assumptions about the data:

1. You can completely ignore the panel dimension and treat the data as a repeated cross-section (i.e., you can ignore the fact that observations from the same individual will be correlated due to unobserved heterogeneity, etc.).
2. You can ignore any issues that often arise when working with time series data such as the macroeconomic and financial variables that you will be using.
3. You can ignore the survey sampling weights if you want. While the weights are required to make the survey representative for the US, not all `scikit-learn` classes support them.

Part 1 — Data preprocessing (SCE)

To load and process the data, perform the following tasks:

1. Import and merge the data contained in the `sce_extract_YYYY.csv` files.
2. Forward-fill any variables that are asked only in the initial wave if you want to use them in your analysis.
3. Drop all survey waves for the year 2025 to avoid the volatility generated by the second Trump administration.
4. Drop all observations with missing data for each variable that you are using in your analysis.
5. Outliers: For the four expectations variables `infl_1y`, `house_price_change`, `prob_unrate_up`, and `prob_stocks_up`, compute the 1st (P1) and 99th (P99) percentiles and keep only observations for which the values are $>$ P1 and $<$ P99.

6. Create the following indicator variables:
   - `optimist_unrate`: 1 if the respondent states that the probability of an increase in unemployment is less than 50% over the next 12 months, and 0 otherwise.
   - `optimist_stocks`: 1 if the respondent states that the probability of an increase in the stock market is more than 50% over the next 12 months, and 0 otherwise.
   - `optimist_house_price`: 1 if the respondent expects an increase in house prices over the next 12 months, and 0 otherwise.

Whenever you drop observations as part of the data cleaning process, report the number of observations dropped at each step.

For both the initial sample and the final sample, report the number of individuals, the number of observations, the number of survey waves, and the first and last survey dates.

Note: While you are free to use any variables from the SCE, you shouldn't choose ones with too many missing values, as this will reduce your sample size. Your final sample should have around 150,000 observations.

Part 2 — Data preprocessing (macro/finance)

Download and save the following macro/finance time series in the `data/` folder:

Yahoo Finance

- WTI Crude Oil Futures (`CL=F`)
- S&P 500 Index (`^GSPC`)

FRED

- Consumer Price Index for All Urban Consumers: All Items in U.S. City Average (`CPIAUCSL`)
- All-Transactions House Price Index for the United States (`USSTHPI`)
- Federal Funds Effective Rate (`FEDFUNDS`)
- Unemployment Rate (`UNRATE`)
- Real Gross Domestic Product (`GDPC1`)
- Nominal Broad U.S. Dollar Index (`DTWEXBGS`)
- 30-Year Fixed Rate Mortgage Average in the United States (`MORTGAGE30US`)

You can do this programmatically in Python (e.g., using `yfinance`) or download the data manually from the websites. However, make sure that all the data files are saved in the `data/` folder and submitted as part of the solution. Your solution code *should not* attempt to download this data on demand.

You will use past observations of these time series to predict household expectations about the next 12 months as observed in the SCE. You have to make sure that you are only using information that was available at the survey date. To this end, resample all daily and weekly data to monthly frequency (using the last observation in a month) and merge them into the SCE dataset based on the date.

To illustrate, imagine that a respondent was surveyed on May 17. You need to merge the time series data as follows:

- For daily, weekly, or monthly data, use the last observation available on April 30.
- For quarterly data, use the observation available on March 31 (the end of the first quarter).

You can also create absolute or relative differences of these time series and merge them into the SCE dataset using the same rules. For example, for 1-month changes, assuming that the respondent was interviewed on May 17,

- For daily, weekly, or monthly data, you can use the 1-month absolute or relative difference observed between March 31 and April 30, or the 12-month difference for the 12 months ending on April 30.
- For quarterly data, you can use the 1-quarter absolute or relative difference between December 31 and March 31, or the 4-quarter difference ending on March 31.

Provide a brief motivation for using these variables in levels and/or as relative or absolute differences.

Part 3 — Exploratory data analysis (SCE)

To get an overview of the expectations variables `infl_1y`, `house_price_change`, `prob_unrate_up`, and `prob_stocks_up`, create the following graphs:

1. For each variable, create a histogram of the responses.
2. For each variable, plot a time series showing its average value over the survey period.

For each of the `optimist_*` variables you created above, plot a time series showing the fraction of optimists over the survey period.

Part 4 — Exploratory data analysis (macro/finance)

To get an overview of how the macro/finance variables evolved over the SCE sample period, create the following graphs:

1. A figure showing the time series of the original variables in levels.
2. A figure showing any variables in relative or absolute differences that you created from the original time series.

Restrict your plots to the survey period of the SCE.

You are also interested in knowing which of these macro/finance variables are correlated with the expectations variables in the SCE. For each of the expectations variables `infl_1y`, `house_price_change`, `prob_unrate_up`, and `prob_stocks_up`, create a table showing the correlation with all macro/finance variables that you are using, sorted so that the most highly correlated macro/finance variables are at the top.

Part 5 — Predicting continuous expectation variables

With the dataset in hand, perform the following tasks:

1. Split the data into a training set (all observations prior to 2024-01-01) and a test set (all observations for the period from 2024-01-01 to 2024-12-31).
2. Fit a series of models to the training data. You have to use at least linear regression, Ridge regression, and Lasso, but you are free to explore any other models implemented in `scikit-learn` (for example `DecisionTreeRegressor`).

   For each model:

   - State whether the model has any hyperparameters or requires feature scaling.
   - Use cross-validation to determine the hyperparameters (or a subset of them), if applicable.
   - For Ridge and Lasso, plot the validation curves showing how the scoring metric (e.g., MSE) changes with the penalty strength.
   - Report the optimal cross-validated hyperparameter and the associated scoring metric on the validation sample as returned by the various cross-validation classes, if applicable.

- Plot the time series showing the *average* actual and predicted values of the expectations variable over the SCE survey period. Clearly indicate the train/test sample split in the figure.
- Report the RMSE and R^2 on the test sample.

Perform the above analysis for each of the four expectations variables `infl_1y`, `house_price_change`, `prob_unrate_up`, and `prob_stocks_up` (note that some models such as `Ridge` can fit multiple targets at once).

For each expectation variable and each model, you should perform the analysis in two steps:

1. Fit the model using only the macro/finance variables as features.
2. Include any other individual-level variables from the SCE in the feature set that you think are useful for improving the predictive power of the model. Comment on whether including these variables leads to a substantial improvement.

   Hints:

   - It is usually a bad idea to simply include all available variables in these models, or to include too many interactions, or to create too many polynomial features from these variables. You should add only a few variables at a time.
   - Check lecture 10 on how to deal with categorical data. You shouldn't treat such variables as continuous.

Part 6 — Predicting binary optimism variables

Repeat the analysis from Part 5 using the binary variables `optimist_unrate`, `optimist_stocks`, and `optimist_house_price` as targets.

Since these are binary variables, this is a classification task. You should at least use logistic regression and Random forest, but you are free to experiment with any other classification models implemented in `scikit-learn`.

For each model:

- Use cross-validation to determine the hyperparameters (or a subset of them), if applicable.
- Report the optimal cross-validated hyperparameter and the accuracy metrics we covered in the lecture on the training sample.
- Plot the time series showing the actual and predicted fractions of optimists over the SCE survey period. Clearly indicate the train/test sample split in the figure.
- Report the accuracy metrics we covered in the lecture on the test sample.

As in Part 5, start by fitting the models using only the macro/finance variables. In a second step, explore whether it is possible to improve the accuracy of the predictions by including individual-level variables from the SCE.

Part 7 — Conclusion

Write a conclusion summarizing your efforts to predict the expectations variables in the SCE. Which models and feature sets performed best?

Your summary should include tables comparing the relevant metrics across the models that you estimated.

Performance issues

With `scikit-learn`, it is easy to run into performance issues, for example when doing cross-validation. If you are unable to complete some tasks because they take too long or because you run out of memory, try the following:

- The dataset is quite large and has many variables that you are unlikely to use. To preserve memory, import only those variables that are required for your analysis.
- Temporarily draw a subsample of the data while you are working on the project to reduce computation time. Once you are satisfied with your implementation, you can revert to the full sample. If you create a smaller subsample, make sure to stratify by calendar year so that you have a sufficiently large number of observations from each year.
- Use 5 folds for cross-validation instead of 10.
- Reduce the number of candidate values considered in cross-validation.
- For `LassoCV` and `ElasticNetCV`, experiment with the setting `selection='random'` instead of the default value, as this might lead to faster convergence.
- Use fewer features (e.g., fewer interactions, lower maximum polynomial degrees, or fewer categorical variables encoded as dummies).
- Use simpler categorical variables with fewer possible values that retain most of the information, e.g., create an indicator `college` instead of using the full categorical variable `educ`.
- Set the `n_jobs` argument accepted by many `scikit-learn` functions according to your hardware:
  - `n_jobs=K` executes `K` tasks in parallel, which speeds up computation. `K` should not be larger than the number of cores on your CPU, e.g., `K=4` or `K=8` for most modern hardware.
  - `n_jobs=-1` uses *all* your CPU cores, if possible.

  However, larger values of `K` increase memory consumption considerably, so if you run out of memory (RAM), you need to lower `K`.

Appendix: Data description

Variables

Legend for variable type:

`C` – continuous; `I` – indicator (0 or 1); `*` – Categorical

Variable	Type	Description
Meta-variables		
`userid`	—	Unique respondent identifier
`wid`	—	Unique wave identifier (date of interview in YYYYMM format)
`date`	C	Date of the survey interview
`tenure`	C	Number of waves the respondent has participated in
`weight`	C	Survey weight
Financial Well-Being		
`financial_past_12m`	`*`	Q1: Do you think you (and any family living with you) are financially better or worse off these days than you were 12 months ago? (1=Much worse off, 2=Somewhat worse off, 3=About the same, 4=Somewhat better off, 5=Much better off)

Variable	Type	Description
financial_12m	*	Q2: And looking ahead, do you think you (and any family living with you) will be financially better or worse off 12 months from now than you are these days? (1=Much worse off, 2=Somewhat worse off, 3=About the same, 4=Somewhat better off, 5=Much better off)
credit_cond_past_12m	*	Q28: Compared to 12 months ago, do you think it is generally harder or easier these days for people to obtain credit or loans? (1=Much harder, 2=Somewhat harder, 3=Equally easy/hard, 4=Somewhat easier, 5=Much easier)
credit_cond_12m	*	Q29: And looking ahead, do you think that 12 months from now it will generally be harder or easier for people to obtain credit or loans than it is these days? (1=Much harder, 2=Somewhat harder, 3=Equally easy/hard, 4=Somewhat easier, 5=Much easier)
prob_miss_paym_3m	C	Q30new: What do you think is the percent chance that, over the next 3 months, you will NOT be able to make one of your debt payments? (0-100%)
Inflation Expectations		
infl_1y	C	Q8v2part2: Expected rate of inflation/deflation over the next 12 months (in percent)
infl_1y_bin_mean	C	Mean of the 1-year-ahead inflation density forecast (from Q9 bins)
infl_1y_bin_var	C	Variance of the 1-year-ahead inflation density forecast (from Q9 bins)
infl_1y_bin_median	C	Median of the 1-year-ahead inflation density forecast (from Q9 bins)
infl_1y_bin_iqr	C	Interquartile range of the 1-year-ahead inflation density forecast (from Q9 bins)
infl_1y_bin_prob_defl	C	Probability of deflation implied by the 1-year-ahead inflation density forecast (0-100)
infl_3y	C	Q9bv2part2: Expected rate of inflation/deflation over the 12-month period between 24 and 36 months from now (in percent)
infl_3y_bin_mean	C	Mean of the 3-year-ahead inflation density forecast (from Q9c bins)
infl_3y_bin_var	C	Variance of the 3-year-ahead inflation density forecast (from Q9c bins)
infl_3y_bin_median	C	Median of the 3-year-ahead inflation density forecast (from Q9c bins)
infl_3y_bin_iqr	C	Interquartile range of the 3-year-ahead inflation density forecast (from Q9c bins)
infl_3y_bin_prob_defl	C	Probability of deflation implied by the 3-year-ahead inflation density forecast (0-100)
infl_5y	C	Q1Apart2: Expected rate of inflation/deflation over the 12-month period between 48 and 60 months from now (in percent)
Housing & Economy		
prob_move_house	C	Q3: What do you think is the percent chance that, over the next 12 months, you will move to a different primary residence? (0-100%)
prob_unrate_up	C	Q4new: What do you think is the percent chance that 12 months from now the unemployment rate in the U.S. will be higher than it is now? (0-100%)
prob_irate_up	C	Q5new: What do you think is the percent chance that 12 months from now the average interest rate on savings accounts will be higher than it is now? (0-100%)
prob_stocks_up	C	Q6new: What do you think is the percent chance that 12 months from now, on average, stock prices in the U.S. stock market will be higher than they are now? (0-100%)
house_price_change	C	Q31v2part2: Expected percent change in the average home price nationwide over the next 12 months
house_price_change_3y	C	C2part2: Expected percent change in the average home price nationwide between 24 and 36 months from now
govt_debt_change	C	C3part2: Expected percent change in the level of U.S. government debt over the next 12 months
Employment & Earnings		

Variable	Type	Description
working	I	Q10: Indicator of whether the respondent is working (full-time or part-time)
num_jobs	C	Q11: Altogether, how many jobs do you have?
self_employed	I	Q12new: Indicator of whether the respondent is self-employed in their current/main job
prob_lose_job	C	Q13new: Percent chance that you will lose your job during the next 12 months (0-100%)
prob_leave_job	C	Q14new: Percent chance that you will leave your job voluntarily during the next 12 months (0-100%)
looking_for_job	I	Q15: Indicator of whether the respondent is currently looking for a job (conditional on not working)
prob_accept_job_12m	C	Q17new: Percent chance that, within the coming 12 months, you will find a job that you will accept (0-100%)
prob_accept_job_3m	C	Q18new: Percent chance that, within the coming 3 months, you will find a job that you will accept (0-100%)
jobless_length	C	Q16/Q19: How long have you been unemployed / out of work (in months)
prob_search_job_12m	C	Q20new: Percent chance that, within the coming 12 months, you will start looking for a job (0-100%)
prob_search_job_3m	C	Q21new: Percent chance that, within the coming 3 months, you will start looking for a job (0-100%)
earnings_change	C	Q23v2part2: Expected percent change in earnings on the main job over the next 12 months
same_employer	I	DSAME: Indicator of whether the respondent works for the same employer as in the previous survey
Household & Income		
hh_inc_change	C	Q25v2part2: Expected percent change in total household income over the next 12 months
hh_spending_change	C	Q26v2part2: Expected percent change in total household spending over the next 12 months
taxes_change	C	Q27v2part2: Expected percent change in total taxes to be paid 12 months from now
hh_inc_bin	*	Q47/D6: Total combined pre-tax income of all members of your household during the past 12 months (11 categories)
hh_inc_bin_rank	C	Median income percentile rank implied by hh_inc_bin in the corresponding calendar year, computed from the American Community Survey (ACS)
owner	I	Q43: Indicator of whether the respondent owns their current primary residence (asked only in the first interview)
couple	I	Q38: Indicator of whether the respondent is married or living with a partner
num_kids	C	Number of children in the household (derived from Q45new/D2new categories)
hh_changed	I	D1: Indicator of whether household composition has changed since the last survey
Demographics & Literacy		
age_init	C	Q32: Respondent's age (recorded at the first interview and carried forward across waves)
female	I	Q33: Indicator of whether the respondent is female
hispanic	I	Q34: Indicator of whether the respondent is of Hispanic, Latino, or Spanish origin
black	I	Q35: Indicator of whether the respondent identifies as Black or African American
educ	*	Q36: Coarse education level (1=LT HS, 2=HS, 3=Some college, 4=College degree)

Variable	Type	Description
health	*	Q45b: Self-reported health (1=Excellent, 2=Very good, 3=Good, 4=Fair, 5=Poor) (asked only in the first interview)
take_fin_risk	*	QRA1: Willingness to take risks regarding financial matters (1=Not willing at all, 5=Very willing)
Numerical Literacy		
num_lit_q1	C	QNUM1: Numerical literacy question (sofa price at a half-price sale)
num_lit_q1_correct	I	Indicator of whether the respondent correctly answered QNUM1
num_lit_q2	C	QNUM2: Numerical literacy question (savings account interest)
num_lit_q2_correct	I	Indicator of whether the respondent correctly answered QNUM2
num_lit_q3	C	QNUM3: Numerical literacy question (lottery winners)
num_lit_q3_correct	I	Indicator of whether the respondent correctly answered QNUM3
num_lit_q5	C	QNUM5: Numerical literacy question (disease prevalence)
num_lit_q5_correct	I	Indicator of whether the respondent correctly answered QNUM5
num_lit_q6	C	QNUM6: Numerical literacy question (viral infection)
num_lit_q6_correct	I	Indicator of whether the respondent correctly answered QNUM6
num_lit_q8	*	QNUM8: Numerical literacy question (interest rate vs. inflation)
num_lit_q8_correct	I	Indicator of whether the respondent correctly answered QNUM8
num_lit_q9	*	QNUM9: Numerical literacy question (stock mutual fund risk)
num_lit_q9_correct	I	Indicator of whether the respondent correctly answered QNUM9

Reference

- The Survey of Consumer Expectations (SCE) is administered by the Federal Reserve Bank of New York.
- URL: https://www.newyorkfed.org/microeconomics/sce#/
- Questionnaire: Link